

Ochrell: Continuous Spectral Reconstruction and Optical-Strength Priors for RGB Pigment Mixing

Ochrell project — independent implementation study

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Abstract

RGB colors do not uniquely identify reflectance spectra or the absorption and scattering coefficients of paint. A useful digital pigment mixer must therefore choose explicit priors as well as reproduce its inputs. We investigate a mathematical revision of an independently implemented Rust Kubelka–Munk mixer, without acquiring pigment measurements or fitting to another engine’s outputs. Three bounded, smooth reflectance anchors and their complements support a continuous linear-RGB reconstruction. A geometric-mean optical normalization assigns separate absorption and scattering strengths; a small linear-light residual restores the source color. The resulting encoder has no nonlinear optimization or lookup table at runtime. On 10000 seeded colors, mean uncorrected reconstruction error falls from 11.5 to 0.072 CIEDE2000. In an independent blue/yellow color-family diagnostic, green-dominant midpoint frequency increases from 38.4% to 80.8%. The fast 41-band model differs from the 81-band double-precision reference by mean 0.00352 and maximum 0.0999 CIEDE2000 over 10000 random mixtures. Median throughput on the recorded host is 1.39 million complete RGB mixtures per second, or 4.37 million with cached optical states. White-tint hue shifts decrease, but latent storage grows and uncached mixing is slower than the preceding LUT implementation. These results establish numerical and design improvements within a synthetic model, not physical accuracy for a particular paint formulation.

1. Introduction

A digital painter expects color mixing to resemble material interaction: yellow and blue should often produce green; adding white should lighten a color with characteristic changes in chroma; complementary colors should lose saturation. Neither encoded RGB averaging nor additive linear-light interpolation models absorption and scattering within paint. A perceptual color space regularizes perceived transitions, but also does not establish a material interpretation.

The main ambiguity arises before mixing. Three RGB components constrain only three integrals of a reflectance spectrum. A spectrum can be selected by regularization, yet opaque reflectance still supplies only the ratio of absorp-

tion to scattering. Different absolute optical strengths can preserve the isolated color and produce different mixtures. Mathematical design can improve a general RGB painting model without new measurements, but cannot identify the actual paint that generated an arbitrary RGB color.

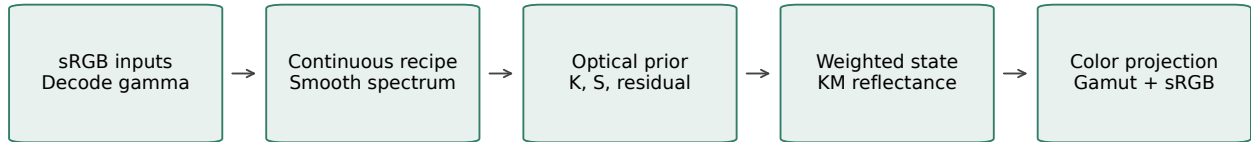
Our preceding implementation used an eight-component synthetic pigment palette, constrained nonlinear inversion, a reconstruction residual and a tetrahedral encoder LUT. Its default yellow/blue example produced green, but additional tests revealed a dark blue/yellow pair that mixed to blue-gray, a muted cyan/magenta mixture and peach-shifting red/white tints. Large residuals and disagreement between interpolated and directly optimized recipes exposed a deeper issue: endpoint accuracy concealed a weak and sometimes unstable material representation.

We compare three responses: refitting the finite pigment palette, directly reconstructing reflectance with conventional scattering assumptions, and reconstructing reflectance with a declared optical-strength prior. The last is selected on recorded diagnostic evidence. This changes the meaning of the basis: it becomes a colorimetric reconstruction basis rather than a palette of identifiable physical ingredients. We retain the old implementation and unsuccessful alternatives so that the change can be examined rather than inferred from favorable images alone.

Our contributions are an independently generated continuous reconstruction and optical-state representation; a dependency-free Rust implementation with separate precision/sampling paths; and a reproducible evaluation of reconstruction, trajectories, tints, sampling, parameter choices and performance. We do not claim invention of spectral upsampling, Kubelka–Munk mixing or residual correction. We do not claim measured paint fidelity, numerical compatibility with Mixbox, or a statistically established preference among artists.

2. Related work

Kubelka and Munk’s two-flux theory and Kubelka’s later development provide approximate relationships between diffuse reflectance, absorption and scattering (Kubelka and Munk 1931; Kubelka 1948). Haase and Meyer apply pigment optics to graphics and paint-oriented color selection (Haase and Meyer 1992). Curtis et al. combine optical lay-



Anchor fitting happens offline. The runtime encoder has no optimizer or LUT.

Figure 1: The default pipeline. Anchor fitting is offline; input-dependent inversion is explicit and continuous.

ers with separate watercolor transport (Curtis et al. 1997). These precedents distinguish homogeneous color mixing from the larger simulation of thickness, substrate and fluid motion.

Spectral reconstruction from RGB is underdetermined. Smits demonstrates compact reflectance bases (Smits 1999); Burns examines smoothness-based reconstruction (Burns 2020); Jakob and Hanika construct an efficient low-dimensional space of bounded spectra (Jakob and Hanika 2019). Our construction follows the broad basis and smoothness motivations, but fits its own spectra using the objective stated below. None of these colorimetric methods, by itself, establishes the independent scattering strength needed for pigment mixing.

Sochorová and Jamriška’s contribution goes beyond applying classical K–M equations: their practical RGB interface uses a latent pigment representation, additive color residuals, surrogate-pigment treatment of gamut limitations and acceleration (Sochorová and Jamriška 2021). Their published representation motivates the compatibility residual used in both of our versions. We do not use their proprietary coefficients, implementation artifacts or lookup tables. The present revision stores wavelength-wise K and S rather than concentrations over our previous finite palette. This avoids that palette’s nonlinear inverse, while increasing state size and giving up ingredient identification.

Pigmento studies pigment-oriented image decomposition and editing, a different problem from a stateless per-color encoder (Tan et al. 2017). RYB-inspired interpolation provides an intuitive color-control alternative without identifying spectra (Gossett and Chen 2004). Neugebauer-type models are appropriate to area coverages and overprinted layers rather than a direct replacement for homogeneous pigment-volume mixing (Hébert and Hersch 2015). We retain sRGB, linear RGB and OKLab (Ottoosson 2020) as inexpensive baselines.

Spectral.js is an open-source RGB pigment-like mixer with a spectral reconstruction and K–M formulation (Spectral.js contributors, n.d.). We compare its public version 3.0.0 API outputs and previously captured Mixbox demonstration outputs after selecting our model. Similarity to these engines is a descriptive comparison, not a measurement of physical truth. No external mixing implementation,

spectra or output samples enter our fitting objective.

3. Color and optical background

3.1 Colorimetric projection

Let r_j denote reflectance at wavelength λ_j , E_j the D65 illuminant and $\bar{x}_j, \bar{y}_j, \bar{z}_j$ the CIE 1931 two-degree color-matching functions. Trapezoidal quadrature gives

$$[X, Y, Z]^T = k \sum_j h_j E_j r_j [\bar{x}_j, \bar{y}_j, \bar{z}_j]^T,$$

where h_j includes interval and endpoint weights and $k^{-1} = \sum_j h_j E_j \bar{y}_j$. The reference uses 380–780 nm at 5 nm intervals. The input tables are the attributed, modified Colour-package subset documented in the repository, rather than an unmodified copy of current official CIE files (Colour Developers, n.d.; International Commission on Illumination 2019b, 2019a).

A fixed XYZ-to-linear-sRGB matrix gives a projection $A \in \mathbb{R}^{3 \times N}$. Each RGB quadrature column is additionally normalized to make a unit reflector project exactly to $[1, 1, 1]$. This numerical white correction is part of our model. It should not be confused with exact unmodified standard colorimetry. The normalized Y quadrature is denoted y_j .

For encoded sRGB channel u , linear intensity is $x = u/12.92$ below or at 0.04045, and $x = ((u + 0.055)/1.055)^{2.4}$ otherwise (World Wide Web Consortium, n.d.). Reflectance, optical mixing and residual addition occur before display encoding. Inputs are finite bounded sRGB, not HDR emission or premultiplied colors.

3.2 Infinite-thickness K–M

With absorption $K_j \geq 0$ and scattering $S_j > 0$, define $q_j = K_j/S_j$. The opaque reflectance and its inverse are

$$r_j = \frac{1}{1 + q_j + \sqrt{q_j(q_j + 2)}}, \quad q_j = \frac{(1 - r_j)^2}{2r_j}.$$

The reciprocal form avoids subtractive cancellation in the usual expression $1+q-\sqrt{q^2+2q}$. We assume linear mixing of K and S with normalized material amounts. This is a selected homogeneous diffuse model; interactions between particles, binder changes and surface reflections are outside it.

Multiplying both K and S of one component by a positive scalar preserves its isolated reflectance but changes its influence in mixtures. Therefore an RGB-only mixer must select both a spectral metamer and an optical-strength convention. Exact endpoint reconstruction cannot resolve this ambiguity.

4. Method

4.1 Offline spectral basis

We fit only the three linear-sRGB primary anchors $p \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Let $\epsilon = 0.001$, $\sigma(z) = 1/(1 + \exp(-z))$ and D be the unscaled first-difference operator on the reference grid. Solve

$$\min_z \|A[\epsilon + (1 - 2\epsilon)\sigma(z)] - p\|_2^2 + \lambda^2 \|Dz\|_2^2,$$

with $\lambda = 0.001$. The logistic parameterization bounds reflectance strictly away from singular black and white. Logit smoothing encourages transitions across wavelength without prescribing measured pigment absorption bands. The reflectance floor limits achievable anchor accuracy; the residual addresses the remaining error.

Initialization is a bounded linear least-squares reflectance fit with a 0.01 first-difference penalty. Its normalized reflectance is clipped to $[10^{-4}, 1 - 10^{-4}]$ before conversion to logit coordinates. SciPy least squares uses an analytic Jacobian, termination tolerances 10^{-12} and at most 2,000 evaluations. Convergence records and a generated-basis checksum are saved. We make no claim of a unique spectrum or globally optimal nonlinear solution.

Given fitted primary spectra B_R, B_G, B_B , define complements $B_C = 1 - B_R$, $B_M = 1 - B_G$, $B_Y = 1 - B_B$, plus constant $B_W = 1 - \epsilon$ and $B_K = \epsilon$. These are eight reconstruction functions generated from three solves. Their coefficients are independent of other mixing libraries. Figure 2 shows their shapes and one inferred optical state.

4.2 Explicit continuous reconstruction

For linear RGB x , let $l \leq m \leq h$ be its sorted channel values. Place weight l on white, $1 - h$ on black, $m - l$ on the secondary complementary to the minimum channel, and $h - m$ on the maximum-channel primary. The four weights are nonnegative and sum to one. Reconstruct $r = \sum_i w_i B_i$.

This is a piecewise-affine cube decomposition. At a channel-order boundary, the switching term has zero weight or

agrees with the limiting recipe, so the spectrum is continuous. The first derivative need not be continuous. Since all anchors lie in $[\epsilon, 1 - \epsilon]$, every reconstructed spectrum satisfies the same bounds. There is no input-dependent iterative optimization and no encoder LUT. The quantities w_i are reconstruction weights, not asserted physical pigment concentrations.

4.3 Optical-strength prior and white behavior

After reconstructing r , calculate q using K–M inversion and $Y = \sum_j y_j r_j$. Define neutral scattering normalization $s_n = 2Y/(1 + Y^2)$. For a neutral spectrum, this equals $1/(1 + q)$. Our explored family is

$$S_j = \eta(1 + q_j)^{-\beta} s_n^{1-\beta}, \quad K_j = q_j S_j,$$

$$\eta = 1 + (w_{\text{white}} - 1) \min(x).$$

At $\beta = 0$, S is wavelength-independent and adjusted to the spectrum’s luminance. At $\beta = 1$, $K_j + S_j = \eta$ independently at every wavelength. We choose $\beta = 1/2$, the geometric mean of these normalizations, and $w_{\text{white}} = 2$. The implementation needs a square root rather than a general fractional power.

The scalar η increases optical strength with the color’s neutral-white component without changing its isolated K/S ratio. This is a declared tinting prior. It is not a measured titanium-white scattering coefficient, and the same RGB color from two different real paints need not follow it. Parameter experiments below show why neither constant S nor a pure per-band budget was adequate for our diagnostics.

4.4 Persistent optical states and residuals

Encode the state $z = (K, S, e)$, where $e = x - Ar$ is a linear-RGB residual. For normalized nonnegative amounts α_i , blend all components:

$$K' = \sum_i \alpha_i K_i, \quad S' = \sum_i \alpha_i S_i, \quad e' = \sum_i \alpha_i e_i.$$

Apply K–M reflectance to K'/S' and return $Ar' + e'$ before gamut mapping. At a source endpoint, this reproduces x algebraically. Public pair mixing additionally returns the exact supplied endpoint objects. Finite-precision encode/decode reconstruction remains a separate test. The small residual does not certify that the inferred spectrum is physically correct.

Weighted optical-state mixtures are associative up to rounding when aggregate weights carry the total amount of each group. Decoding a mixture to RGB and re-encoding it

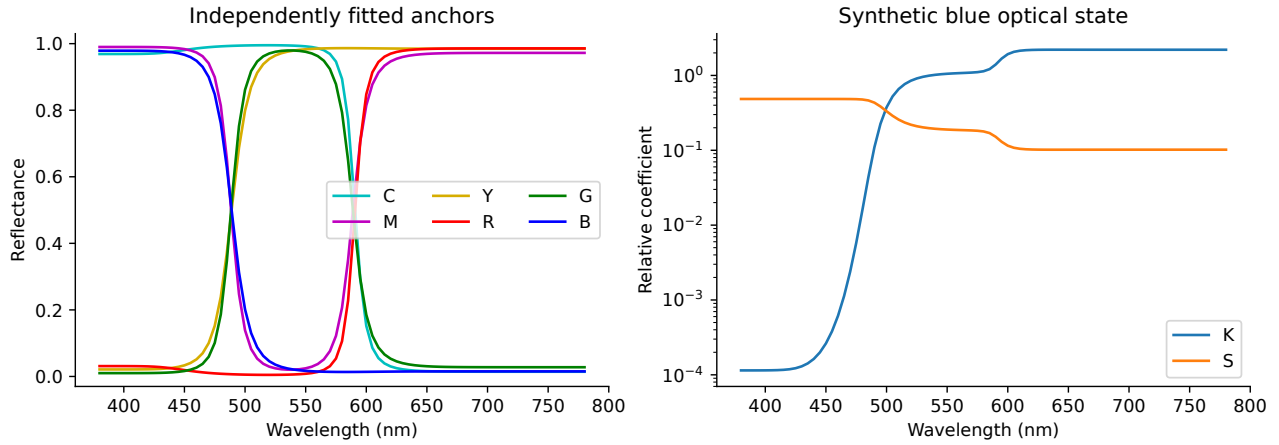


Figure 2: Fitted reflectance anchors and the inferred K/S state for sRGB (20,70,255). The spectra are mathematical priors, not pigment measurements.

loses spectral information and is not associative. This distinction matters for brush engines: repeatedly mixing only display colors does not retain the previous material state.

4.5 Gamut mapping and continuity

We retain the earlier neutral-axis gamut compression. Compute $L = \text{clamp}(0.2126x_r + 0.7152x_g + 0.0722x_b, 0, 1)$ and select the largest $a \in [0, 1]$ placing $L + a(x - L)$ inside the RGB cube. The map is identity in gamut and preserves this luminance when it is in range; it does not preserve perceptual hue exactly.

All ideal-arithmetic stages are continuous on the bounded input domain. Strictly positive scattering prevents division singularities. This establishes continuity, not constant perceptual speed, monotone hue or monotone chroma. In particular, OKLab’s cube root is highly sensitive near black. Numerical trajectories and endpoint refinement are reported separately from the structural argument.

5. Reference and fast implementation

The reference uses 81 wavelength samples at 5 nm and f64 optical arithmetic. The fast path uses 41 samples at 10 nm and f32 arrays, with f64 transfer functions, residual addition and final gamut mapping. A shared Rust kernel keeps their equations aligned; an independent NumPy implementation provides an additional check. We report both spectral-discretization error and same-grid arithmetic error.

The default mixer is stateless and zero-sized. Encoding, pair mixing and weighted accumulation require no heap allocation or platform-specific calls. Weighted accumulation uses f64 before storing the result at the selected precision. The crate forbids unsafe code and has no Cargo dependencies. Python, SciPy, Colour and Matplotlib are offline fitting/evaluation tools. The core design is compatible

with a WASM integration path; browser performance is not measured here.

A fast optical state occupies 340 bytes, versus 44 for the legacy concentration-plus-residual state. The new basis and fast quadrature occupy approximately 1968 bytes as numeric arrays; the default no longer parses the legacy 1149992-byte LUT. Legacy assets remain in the research repository. Storing a state per canvas pixel is consequently more expensive even though startup and encoder-table memory are reduced. Brush-color caching or sparse material storage may be preferable.

The final encoder integrates its reconstructed reflectance directly when computing the residual. In exact arithmetic, subsequent K–M inversion recovers that reflectance. This removes redundant square roots at the cost of small measured f32 roundoff. The optimized implementation retains exact public pair endpoints and validates the encode/decode round trip. Explicit SIMD, GPU execution and compression are future work.

6. Experimental methodology

We use 10000 uniformly sampled encoded-sRGB colors and 10000 random pairs with uniformly sampled t , generated by the recorded xorshift64 sequence with seed 20260930. Uniform encoded RGB is not uniform perceptual color or a model of artists’ usage. The same samples compare the old default with the new default. Reconstruction is evaluated both with and without residual correction using CIEDE2000 (Sharma, Wu, and Dalal 2005). OKLab distance multiplied by 100 is used for local trajectory diagnostics; it is not interchangeable with CIEDE2000.

Thirteen canonical pairs include ordinary and saturated primaries, complements, white and black, and the dark-blue/yellow failure case from the earlier comparison. Each has 1001 samples. We also evaluate 200 random trajectories,

5,000 input perturbations of 10^{-5} in one encoded channel, and 1,024 white tints at the same dense spacing. Tint hue shifts are compared only where source chroma exceeds 0.04 and both models’ midpoint chroma exceeds 0.02, giving 957 common eligible cases. Hue retention is a design diagnostic, not a general physical law.

Parameter selection used 512 blue/yellow pairs and 1,024 white-tint colors generated independently in NumPy with seed 20260930. Yellow hues span 45–75 degrees, blue hues 220–255 degrees; saturation spans 0.6–0.95, yellow value 0.5–1 and blue value 0.2–1. A held-out 2,048-pair family uses seed 20261001. We call a midpoint green-dominant when its green channel exceeds red and blue and its OKLab chroma is at least 0.03. This is a deliberately simple algebraic criterion. Not every real pigment pair in these RGB ranges would necessarily satisfy it.

Timings use 7 repetitions of 100000 operations, varying among 256 seeded colors and 997 interpolation values. Rust `black_box` prevents constant folding. Runs are single-threaded, release mode, LTO disabled and one codegen unit, without `target-cpu=native`. The host reports AMD EPYC 9V74 80-Core Processor; compiler is `rustc 1.75.0 (82e1608df 2023-12-21)` (built from a source tarball). Raw repetitions and full environment are stored. Cached timings include interpolation and display decoding but exclude the two input encodes. CPU frequency, cache conditions and shared-host load limit transferability.

6.1 External comparison protocol

The external comparison is deliberately frozen. Mixbox observations are median pixels from the official gradient demo’s JPEG captures at $t=0.25, 0.5$ and 0.75 for the same thirteen pairs. The 39-point sRGB control has mean approximately 0.286 and maximum 0.804 CIEDE2000 capture discrepancy. Mixbox comparisons are therefore approximate observations, not SDK equality tests.

Spectral.js 3.0.0 supplies 401 public-API samples per pair, using default tinting strength 1 and gamut method `map`; outputs are its native 8-bit sRGB values. Its version, package checksum, invocation and endpoint checks are frozen with the data. We use no external spectra, coefficients or implementation internals, and neither comparator enters candidate fitting or selection. The bundled baseline comparison archive supports reanalysis of the observations; reacquiring a future live demo would be a new experiment.

7. Results

7.1 Reconstruction and approximation

The main improvement is in the uncorrected representation: mean error decreases from 11.5 to 0.072 CIEDE2000 on the same colors. The large legacy residual was hiding substantial palette mismatch. The new residual is smaller because its anchors directly target colorimetric reconstruction. The

worst uncorrected new error is 0.85. Dark colors remain difficult because the positive reflectance floor prevents true zero reflectance.

Corrected mean error is $1.27\text{e-}06$, with maximum $1.79\text{e-}05$. These values mostly measure algebra and floating-point roundoff. They should not be read as exceptionally accurate pigment identification. A separate 16^3 grid including RGB-cube faces has maximum encoded-channel error $1.98\text{e-}05$ and maximum $6.79\text{e-}05$ CIEDE2000; gamma and gamut compression amplify small optical roundoff at cube faces. All grid colors retain their exact 8-bit values. The full f64 Rust output agrees with the independent Python reference to maximum linear-channel difference $6.64\text{e-}08$. On the fast sampling grid, f32/Python mixture disagreement has mean $5.83\text{e-}06$ CIEDE2000. Promoting the fast decoder alone to f64 produces a maximum linear-channel difference of $3.44\text{e-}07$.

The new fast/reference mixture error has mean 0.00352, P95 0.013 and maximum 0.0999 CIEDE2000. The 10 nm choice substantially reduces the 20 nm and 40 nm outliers. A 1 nm evaluation of the interpolated fitted spectra differs from the 5 nm reference by mean 0.026 and maximum 0.268. This control prevents treating the chosen reference grid as an exact continuum solution.

Only 0.46% of the new random mixtures require gamut compression. Their unmapped channel range is $[-0.01698, 1.00485]$. This fraction measures an implementation intervention, not whether an in-gamut result is physically correct.

7.2 Mixing and tint behavior

The dark blue/yellow failure is resolved under the green-dominant diagnostic; cyan/magenta becomes distinctly blue-violet and red/white becomes pinker. The full canonical gradients are shown without omitting unsuccessful or less favorable cases. New red/blue is less saturated than the older dark purple, and the pure-yellow/pure-blue midpoint is darker. These are tradeoffs rather than unqualified improvements.

The held-out green-dominant frequency rises from 38.4% to 80.8%. The remaining non-green outcomes show that the rule is not guaranteed for arbitrary blue/yellow inputs. In the common eligible white-tint set, median hue shift decreases from 4.21 to 2.34 degrees, and P95 decreases from 35.6 to 10.4 degrees. Both models show monotone sampled luminance for the tested tints. This supports improved hue retention in the declared diagnostic, not a claim about any named white pigment.

7.3 Smoothness and sensitivity

The maximum adjacent random-trajectory step decreases from 1.59 to 1.17 ΔEOK100 at $\Delta t = 0.001$. The new P99 of per-trajectory maxima is 0.929 versus 0.815 previously,

Table 1: Source reconstruction (CIEDE2000), identical sample set.

Reconstruction	Mean	Median	P95	P99	Max
v0.2 corrected	1.27e-06	4.91e-07	4.84e-06	7.45e-06	1.79e-05
v0.2 raw	0.072	0.0341	0.272	0.517	0.85
v0.1 raw	11.5	9.2	28.6	35.7	42.1

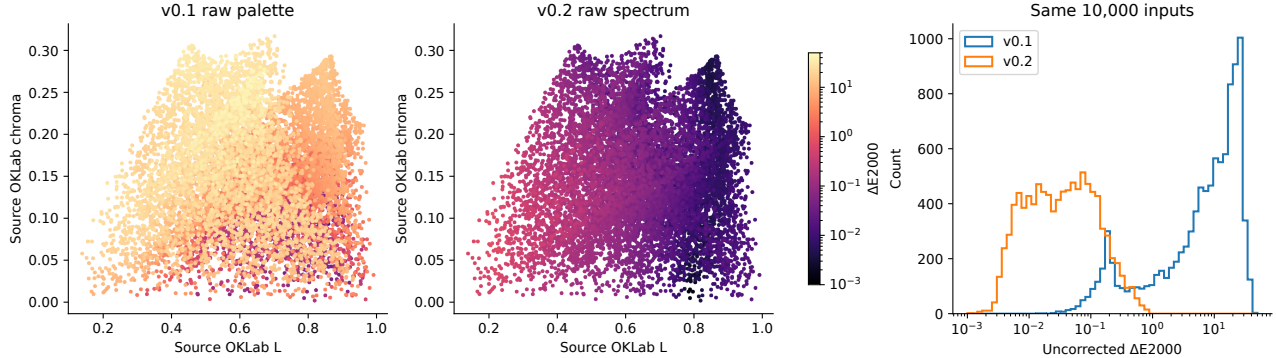


Figure 3: Uncorrected reconstruction on the same 10,000 colors. The shared color scale exposes both the earlier palette’s large errors and the new spectrum’s remaining dark-region errors.

so the tail does not improve uniformly. The largest mid-point response to an encoded-channel perturbation falls from 0.0132 to 0.00169 ΔEOK100 . This is consistent with removing ambiguous inverse recipes, but finite sampling cannot certify a global Lipschitz bound.

Pure black remains a special case. Its first black/white step has 10 ΔEOK100 ; linear RGB interpolation gives 10 under the same sampling. The new trajectory converges toward black as t decreases, and sampled linear luminance increases monotonically. An initial test imposing an arbitrary universal bound of 2 ΔEOK100 per step failed here. We retain that negative finding and separate continuity from perceptual speed. No discontinuity or nonfinite value was observed in the evaluated trajectories.

7.4 Runtime and state cost

The revised fast path processes 1.39 million complete RGB mixtures per second and 4.37 million cached mixtures per second on this host. It is slower than the old default in both modes. Its reference is much cheaper than the preceding optimizer-based reference because reconstruction is explicit; this is an architectural change, not merely a compiler optimization. The larger state is an important integration cost. The measurements support interactive use for many workloads, but not a blanket claim that per-pixel spectral storage or every application is practical.

8. Ablation studies and negative findings

Refitting the original eight-pigment palette with the new reflectances, while retaining finite-palette K/S mixing and nonlinear inversion, did not solve the diagnostic failures. Its half black/white result was [35,35,35], and the dark-blue/yellow midpoint remained teal. This does not prove that no finite palette could work; it shows that improved isolated-color coverage alone was insufficient for the tested construction.

Constant scattering combined with near-black spectra produced excessive absorption in black/white mixing. Conversely, pure per-band normalization ($\beta = 1$) improved some violet mixtures but reduced the frequency of convincing green blue/yellow results. The selection sweep shows the conflict between green-family coverage and tint hue retention. The chosen $\beta = 0.5$ is a compromise; $\beta = 0.25$ retains more green-family results but shifts tint hues more. White strengths 1, 2 and 4 all preserve isolated reflectance, yet produce different tint trajectories. This is a direct demonstration of the optical-strength ambiguity.

Logit smoothness values 0.0001, 0.001, 0.01 and 0.03 were recorded. Larger values improved spectral smoothness at the expense of saturated-anchor fidelity. The selected value was not fit to a target set of real mixtures. Residual removal has a much smaller effect on source accuracy after the revision, but the residual remains useful for bounded black/white and sampling error. Eliminating it entirely would break exact RGB compatibility without establishing physical correctness.

The 20 nm approximation offers fewer samples but larger

Table 2: Sampling discrepancy from the 5 nm computational reference.

Spacing (nm)	Mean	P95	Max
1	0.026	0.0861	0.268
5	0	0	0
10	0.00352	0.013	0.0999
20	0.087	0.341	2.35
40	0.849	2.84	10.5

Table 3: Canonical half-mixture colors.

Pair	v0.1 sRGB8	v0.2 sRGB8
yellow + blue	(49, 144, 112)	(101, 152, 94)
red + blue	(88, 0, 71)	(106, 57, 98)
red + yellow	(242, 118, 14)	(244, 106, 37)
cyan + magenta	(148, 101, 130)	(109, 93, 186)
magenta + yellow	(237, 120, 116)	(236, 105, 83)
black + white	(141, 141, 141)	(166, 166, 166)
red + green	(137, 100, 39)	(116, 93, 53)
blue + orange	(103, 87, 91)	(108, 114, 83)
yellow + purple	(162, 130, 134)	(187, 126, 84)
blue + white	(107, 188, 255)	(135, 169, 255)
red + white	(255, 166, 124)	(250, 131, 139)
extreme + yellow + blue	(60, 174, 124)	(86, 138, 100)
official + default	(73, 97, 107)	(99, 138, 68)

outliers; 40 nm is substantially worse. The 10 nm choice favors accuracy while retaining measured interactive throughput. We did not add a second LUT, neural approximation or explicit SIMD layer because the continuous encoder already met the selected runtime target and avoided an additional approximation to validate.

9. Comparison with other engines

On the 39 frozen Mixbox observations, our mean difference decreases from 13.5 to 6.92 CIEDE2000; the new maximum remains 23.2. Spectral.js differs from those same observations by mean 7.86. Across all 5,213 Spectral.js samples, the old/new differences are 13.1 and 6.49 respectively. These reductions are descriptive and were measured after selection. They do not demonstrate equivalence, establish a ranking against real paint, or justify adapting our coefficients to proprietary outputs.

Mixbox captures have JPEG uncertainty and a small selected sample count. Spectral.js produces quantized 8-bit API outputs under specified settings. The paired endpoints and t values make the experiment useful for visual comparison, while the different acquisition methods limit numerical interpretation. The supplement shows the complete external comparison and all retained metadata. A proper physical validation would instead specify paints, amounts, preparation, thickness, substrate and illumination, then compare held-out measured mixtures.

10. Discussion and limitations

The revision demonstrates that mathematical work can remove large artificial reconstruction errors and improve declared visual diagnostics without new experimental paint data. It cannot determine the correct metamer or strength for an unknown RGB source. Good endpoint colorimetry, green midpoints, hue retention and similarity to other software are different criteria; none substitutes for physical validation.

The new reconstruction basis spans display colors more effectively because it is designed for that purpose. Its spectra and strengths are synthetic and may not correspond to manufacturable pigments. The optical prior favors bounded, tractable behavior and was selected on a limited diagnostic family. Results remain dependent on the observer, illuminant, wavelength interval, reflectance bounds, smoothing and gamut policy. The method does not model real metameric changes under different lights.

Infinite-thickness diffuse optics omit film thickness, translucent glazing, gloss, fluorescence, metallic and interference pigments, granulation, binder-dependent behavior, particle-size effects and spatial transport. These limitations are relevant to the implemented equations, rather than a claim that a different K-M formulation could never address some of them. The positive spectral floor and numerical white normalization are additional modeling approximations.

The reference is a high-precision evaluation of our model, not experimental ground truth. Tests establish finite-sample properties and selected algebraic invariants. There

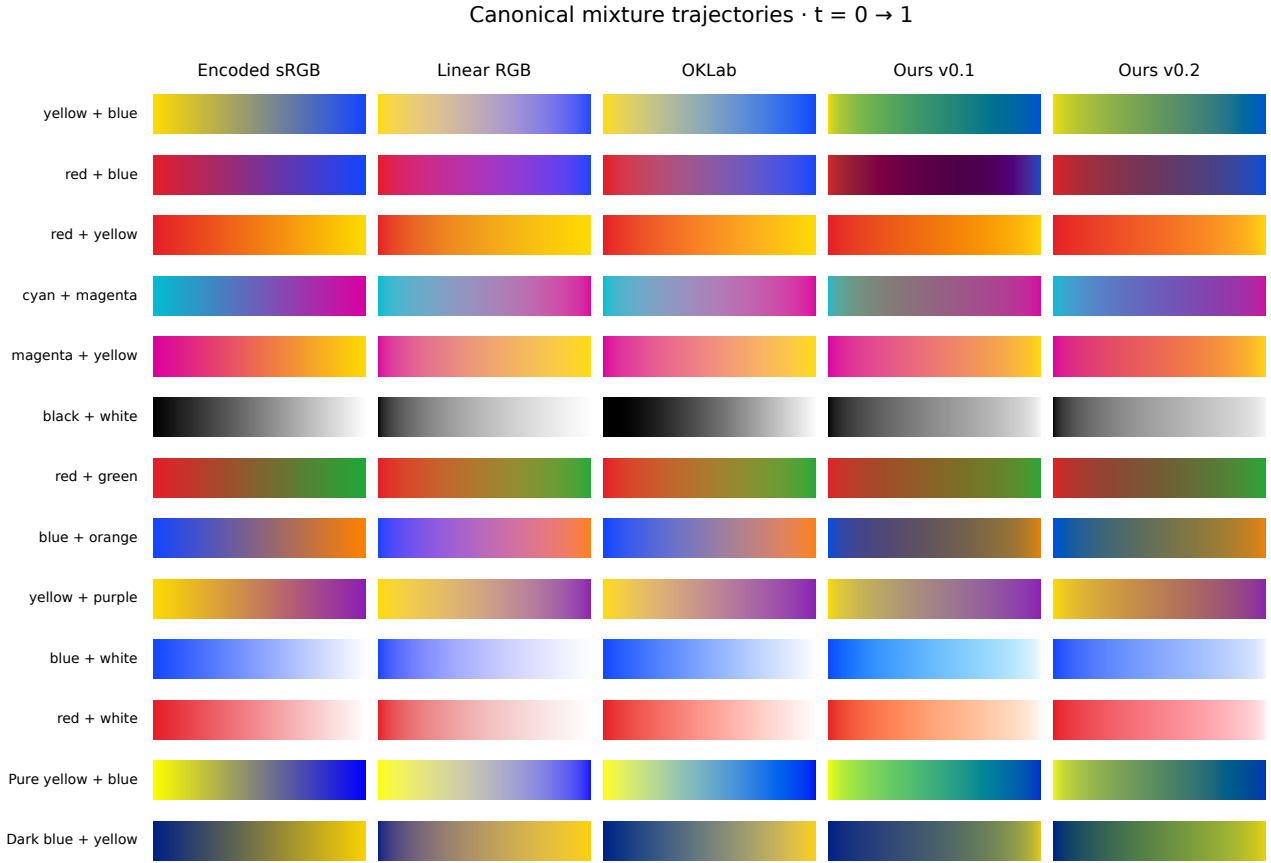


Figure 4: All canonical gradients, with encoded-sRGB, linear-light, perceptual and both pigment models. No panel represents measured paint.

was no human preference study. Reduced tint hue shift is an explicit design preference; real paint can shift hue during tinting. Random encoded-RGB tests overrepresent some color regions and underrepresent actual painting workflows. The state size and slower full-RGB path may matter more than the improved diagnostics in a memory-limited application.

11. Reproducibility and future work

`python3 tools/reproduce.py` reads the root TOML, fits the anchors, builds and tests Rust, reruns the experiments, regenerates figures and tables, and renders the manuscript when Pandoc/TeX are available. The package pins Python dependencies, records compiler/CPU configuration and seeds, and stores raw samples and optimizer results. The legacy implementation, paper and reproduction workflow are preserved. External comparisons use frozen observations by default; they do not silently query a changing live service.

Original code is MIT OR Apache-2.0. Attributed CIE-derived data and derived numerical assets carry CC BY-SA 4.0 notices, so the complete distribution is not exclusively MIT-licensed. The generated basis is inspectable CSV and

Rust arrays, not unexplained binary data. No proprietary Mixbox artifact is required to build, fit or execute the engine.

Useful next work includes state compression validated against the current optical state, explicit SIMD/GPU kernels, WASM execution measurements, artist evaluation and workload-specific caching. Measured paint data, if later available, could define a separate calibrated material mode rather than retroactively making arbitrary RGB inversion unique. Finite-layer optics and substrate effects should be introduced with their own validation instead of adjusting opaque mixing heuristics to mimic glazing.

12. Conclusion

Replacing a constrained finite-palette inverse with continuous spectral reconstruction substantially reduces raw color mismatch and improves the selected blue/yellow and tint diagnostics. The chosen optical-strength normalization is essential: better reflectance reconstruction alone did not provide suitable mixture behavior. The fast implementation retains low error against its own reference and measured interactive throughput, with larger cached state and slower mixing than the previous LUT model. Residual

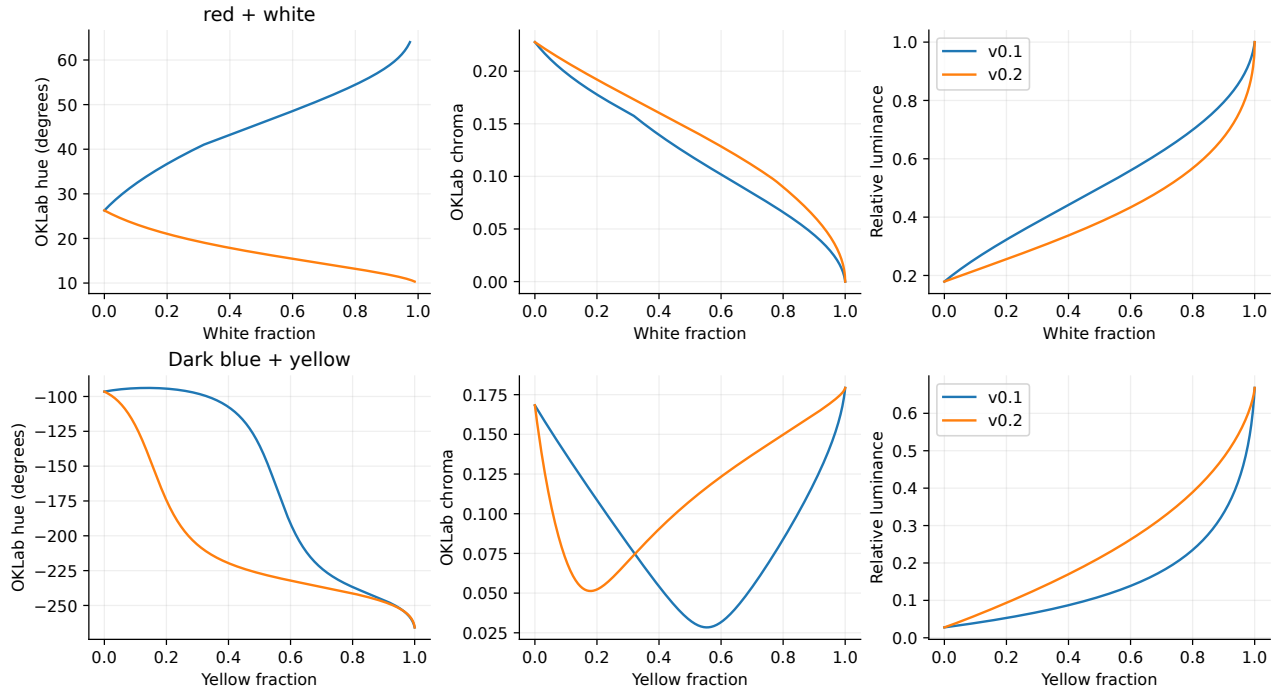


Figure 5: Hue, chroma and luminance for red/white and the dark-blue/yellow failure case. Hue is omitted near neutral colors, where it is unstable.

Table 4: Single-thread runtime; cached states omit input encoding.

Operation	Median ns	Million/s
sRGB	15.6	64
linear RGB	148	6.75
OKLab	272	3.67
v0.1 RGB	572	1.75
v0.2 RGB	719	1.39
v0.2 cached	229	4.37
v0.2 reference	1.87e+03	0.534
v0.1 cached	182	5.5

correction ensures RGB compatibility, while the improved uncorrected results reduce reliance on that correction. The result is a reproducible, physically motivated painting surrogate whose assumptions and remaining failures are explicit; it is not a calibrated predictor of unspecified real paints.

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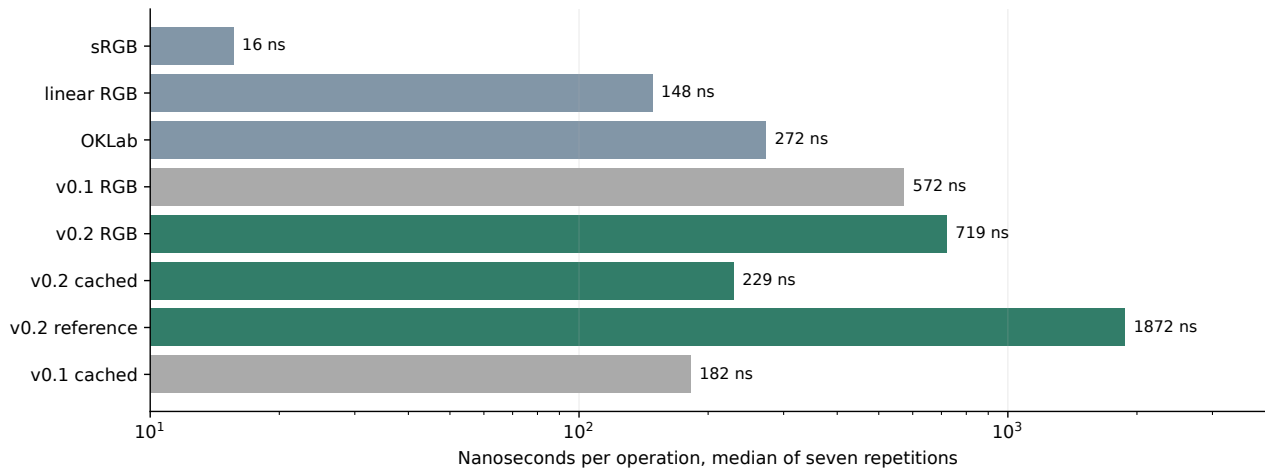


Figure 6: Measured median operation costs. Cached rows omit input encoding; all mixing rows include final display conversion.

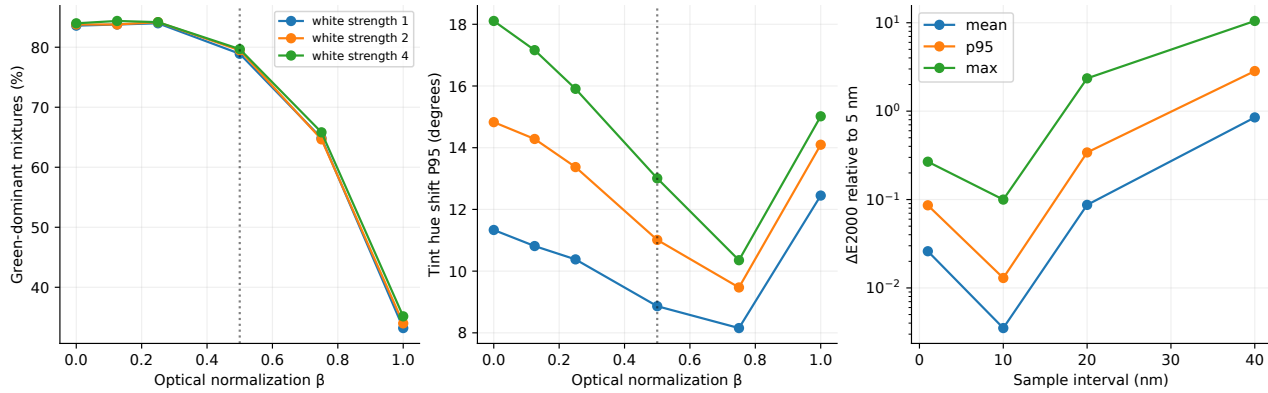


Figure 7: Selection-set normalization and white-strength ablations, plus held-fixed-spectrum sampling error. The dotted line marks the chosen β ; these curves are design diagnostics, not paint-fit errors.

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